1. A motorist drives south at 20.0 m/s for 3.00 min, then turns west and travels at 25.0 m/s for 2.00 min, and finally travels northwest at 30.0 m/s for 1.00 min. For this 6.00-min trip, find (a) the total vector displacement, (b) the average speed, and (c) the average velocity. Let the positive x axis point east.

For each segment of the motion we model the car as a particle under constant velocity. Her displacements are

$$\vec{R} = (20.0\ \text{m/s})(180\ \text{s})\ \textbf{south} + (25.0\ \text{m/s})(120\ \text{s})\ \textbf{west} + (30.0\ \text{m/s})(60.0\ \text{s})\ \textbf{northwest}$$

Choosing $\hat{i}$ = east and $\hat{j}$ = north, we have

$$\vec{R} = (3.60\ \text{km})(-\hat{j}) + (3.00\ \text{km})(-\hat{i}) + (1.80\ \text{km})\cos 45°(-\hat{i}) + (1.80\ \text{km})\sin 45°(\hat{j})$$

$$\vec{R} = (3.00 + 1.27)\ \text{km}(-\hat{i}) + (1.27 - 3.60)\ \text{km}(\hat{j})$$

$$= (-4.27\hat{i} - 2.33\hat{j})\ \text{km}$$

The answer can also be written as

$$\vec{R} = \sqrt{(-4.27\ \text{km})^2 + (-2.33\ \text{km})^2} = 4.87\ \text{km}$$

$$\text{at } \tan^{-1}\left(\frac{2.33}{4.27}\right) = 28.6°$$

or $\boxed{4.87\ \text{km at } 28.6°\ \text{S of W}}$

(b) The total distance or path length traveled is $(3.60 + 3.00 + 1.80)\ \text{km} = 8.40\ \text{km}$, so

$$\text{average speed} = \left(\frac{8.40\ \text{km}}{6.00\ \text{min}}\right)\left(\frac{1.00\ \text{min}}{60.0\ \text{s}}\right)\left(\frac{1000\ \text{m}}{\text{km}}\right) = \boxed{23.3\ \text{m/s}}$$

(c) $$\text{Average velocity} = \frac{4.87 \times 10^3\ \text{m}}{360\ \text{s}} = \boxed{13.5\ \text{m/s along } \vec{R}}$$

3. Suppose the position vector for a particle is given as a function of time by $\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}}$, with $x(t) = at + b$ and $y(t) = ct^2 + d$, where $a = 1.00\ \text{m/s}$, $b = 1.00\ \text{m}$, $c = 0.125\ \text{m/s}^2$, and $d = 1.00\ \text{m}$. (a) Calculate the average velocity during the time interval from $t = 2.00\ \text{s}$ to $t = 4.00\ \text{s}$. (b) Determine the velocity and the speed at $t = 2.00\ \text{s}$.

(a) For the average velocity, we have

$$\vec{\mathbf{v}}_{\text{avg}} = \left(\frac{x(4.00) - x(2.00)}{4.00\ \text{s} - 2.00\ \text{s}}\right)\hat{\mathbf{i}} + \left(\frac{y(4.00) - y(2.00)}{4.00\ \text{s} - 2.00\ \text{s}}\right)\hat{\mathbf{j}}$$

$$= \left(\frac{5.00\ \text{m} - 3.00\ \text{m}}{2.00\ \text{s}}\right)\hat{\mathbf{i}} + \left(\frac{3.00\ \text{m} - 1.50\ \text{m}}{2.00\ \text{s}}\right)\hat{\mathbf{j}}$$

$$\vec{\mathbf{v}}_{\text{avg}} = \boxed{\left(1.00\hat{\mathbf{i}} + 0.750\hat{\mathbf{j}}\right)\ \text{m/s}}$$

(b) For the velocity components, we have

$$v_x = \frac{dx}{dt} = a = 1.00 \text{ m/s}$$

and $$v_y = \frac{dy}{dt} = 2ct = \left(0.250 \text{ m/s}^2\right)t$$

Therefore,

$$\vec{v} = v_x\hat{i} + v_y\hat{j} = (1.00 \text{ m/s})\hat{i} + \left(0.250 \text{ m/s}^2\right)t\hat{j}$$

$$\boxed{\vec{v}(t = 2.00 \text{ s}) = (1.00 \text{ m/s})\hat{i} + (0.500 \text{ m/s})\hat{j}}$$

and the speed is

$$\left|\vec{v}(t = 2.00 \text{ s})\right| = \sqrt{(1.00 \text{ m/s})^2 + (0.500 \text{ m/s})^2} = \boxed{1.12 \text{ m/s}}$$

4. The coordinates of an object moving in the xy plane vary with time according to the equations $x = -5.00 \sin \omega t$ and $y = 4.00 - 5.00 \cos \omega t$, where ω is a constant, x and y are in meters, and t is in seconds. (a) Determine the components of velocity of the object at $t = 0$. (b) Determine the components of acceleration of the object at $t = 0$. (c) Write expressions for the position vector, the velocity vector, and the acceleration vector of the object at any time $t > 0$. (d) Describe the path of the object in an xy plot.

(a) From $x = -5.00 \sin \omega t$, we determine the components of the velocity by taking the time derivatives of x and y:

$$v_x = \frac{dx}{dt} = \left(\frac{d}{dt}\right)(-5.00 \sin \omega t) = -5.00\omega \cos \omega t$$

and $$v_y = \frac{dy}{dt} = \left(\frac{d}{dt}\right)(4.00 - 5.00 \cos \omega t) = 0 + 5.00\omega \sin \omega t$$

At $t = 0$,

$$\vec{v} = (-5.00\omega \cos 0)\hat{i} + (5.00\omega \sin 0)\hat{j} = \boxed{-5.00\omega \hat{i}\ \text{m/s}}$$

(b) Acceleration is the time derivative of the velocity, so

$$a_x = \frac{dv_x}{dt} = \frac{d}{dt}(-5.00\omega\cos\omega t) = +5.00\omega^2 \sin\omega t$$

and $$a_y = \frac{dv_y}{dt} = \left(\frac{d}{dt}\right)(5.00\omega \sin\omega t) = 5.00\omega^2 \cos\omega t$$

At t = 0,

$$\vec{a} = (5.00\omega^2 \sin 0)\hat{i} + (5.00\omega^2 \cos 0)\hat{j} = \boxed{5.00\omega^2\hat{j}\ \text{m/s}^2}$$

(c) $$\vec{r} = x\hat{i} + y\hat{j} = \boxed{(4.00\ \text{m})\hat{j} + (5.00\ \text{m})(-\sin\omega t\hat{i} - \cos\omega t\hat{j})}$$

$$\vec{v} = \boxed{(5.00\ \text{m})\omega\left[-\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}\right]}$$

$$\vec{a} = \boxed{(5.00\ \text{m})\omega^2\left[\sin\omega t\hat{i} + \cos\omega t\hat{j}\right]}$$

(d) the object moves in a circle of radius 5.00 m centered at (0, 4.00 m)

6. **W** A particle initially located at the origin has an acceleration of $\vec{a} = 3.00\hat{j}$ m/s^2 and an initial velocity of $\vec{v}_i = 5.00\hat{i}$ m/s. Find (a) the vector position of the particle at any time t, (b) the velocity of the particle at any time t, (c) the coordinates of the particle at $t = 2.00$ s, and (d) the speed of the particle at $t = 2.00$ s.

$$\vec{a} = 3.00\hat{j}\ \text{m/s}^2;\ \vec{v}_i = 5.00\hat{i}\ \text{m/s};\ \vec{r}_i = 0\hat{i} + 0\hat{j}$$

(a) The position of the particle is given by Equation 4.9:

$$\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2}\vec{a}t^2 = (5.00\ \text{m/s})t\hat{i} + \frac{1}{2}(3.00\ \text{m/s}^2)t^2\hat{j}$$

$$= \boxed{5.00t\hat{i} + 1.50t^2\hat{j}}$$

where r is in m and t in s.

(b) The velocity of the particle is given by Equation 4.8:

$$\vec{v}_f = \vec{v}_i + \vec{a}t = \boxed{5.00\hat{i} + 3.00t\,\hat{j}}$$

where v is in m/s and t in s.

(c) To obtain the particle's position at $t = 2.00$ s, we plug into the equation obtained in part (a):

$$\vec{r}_f = (5.00\ \text{m/s})(2.00\ \text{s})\hat{i} + \left(1.50\ \text{m/s}^2\right)(2.00\ \text{s})^2\hat{j}$$

$$= \left(10.0\hat{i} + 6.00\hat{j}\right)\ \text{m}$$

so $x_f = \boxed{10.0\ \text{m}}$, $y_f = \boxed{6.00\ \text{m}}$

(d) To obtain the particle's speed at $t = 2.00$ s, we plug into the equation obtained in part (b):

$$\vec{v}_f = \vec{v}_i + \vec{a}t = (5.00\ \text{m/s})\hat{i} + \left(3.00\ \text{m/s}^2\right)(2.00\ \text{s})\hat{j}$$

$$= \left(5.00\hat{i} + 6.00\hat{j}\right)\ \text{m/s}$$

$$v_f = \sqrt{v_{xf}^2 + v_{yf}^2} = \sqrt{(5.00\ \text{m/s})^2 + (6.00\ \text{m/s})^2} = \boxed{7.81\ \text{m/s}}$$

7. The vector position of a particle varies in time accord-
W ing to the expression $\vec{\mathbf{r}} = 3.00\hat{\mathbf{i}} - 6.00t^2\hat{\mathbf{j}}$, where $\vec{\mathbf{r}}$ is in meters and t is in seconds. (a) Find an expression for the velocity of the particle as a function of time. (b) Determine the acceleration of the particle as a function of time. (c) Calculate the particle's position and velocity at $t = 1.00$ s.

(a) We differentiate the equation for the vector position of the particle with respect to time to obtain its velocity:

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt} = \left(\frac{d}{dt}\right)\left(3.00\hat{\mathbf{i}} - 6.00t^2\hat{\mathbf{j}}\right) = \boxed{-12.0t\hat{\mathbf{j}}\ \text{m/s}}$$

(b) Differentiating the expression for velocity with respect to time gives the particle's acceleration:

$$\vec{\mathbf{a}} = \frac{d\vec{\mathbf{v}}}{dt} = \left(\frac{d}{dt}\right)\left(-12.0t\hat{\mathbf{j}}\right) = \boxed{-12.0\hat{\mathbf{j}}\ \text{m/s}^2}$$

(c) By substitution, when $t = 1.00$ s,

$$\boxed{\vec{\mathbf{r}} = \left(3.00\hat{\mathbf{i}} - 6.00\hat{\mathbf{j}}\right)\text{m};\ \vec{\mathbf{v}} = -12.0\hat{\mathbf{j}}\ \text{m/s}}$$

9. AMT M A fish swimming in a horizontal plane has velocity $\vec{\mathbf{v}}_i = (4.00\hat{\mathbf{i}} + 1.00\hat{\mathbf{j}})$ m/s at a point in the ocean where the position relative to a certain rock is $\vec{\mathbf{r}}_i = (10.0\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}})$ m. After the fish swims with constant acceleration for 20.0 s, its velocity is $\vec{\mathbf{v}} = (20.0\hat{\mathbf{i}} - 5.00\hat{\mathbf{j}})$ m/s. (a) What are the components of the acceleration of the fish? (b) What is the direction of its acceleration with respect to unit vector $\hat{\mathbf{i}}$? (c) If the fish maintains constant acceleration, where is it at $t = 25.0$ s and in what direction is it moving?

$\vec{\mathbf{v}}_i = \left(4.00\hat{\mathbf{i}} + 1.00\hat{\mathbf{j}}\right)$ m/s and $\vec{\mathbf{r}}_i = (10.00\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}})$ m

At the first "final" point we consider, 20.0 s later,

$$\vec{\mathbf{v}}_f = (20.0\hat{\mathbf{i}} - 5.00\hat{\mathbf{j}}) \text{ m/s}$$

(a) $$a_x = \frac{\Delta v_x}{\Delta t} = \frac{20.0 \text{ m/s} - 4.00 \text{ m/s}}{20.0 \text{ s}} = \boxed{0.800 \text{ m/s}^2}$$

$$a_y = \frac{\Delta v_y}{\Delta t} = \frac{-5.00 \text{ m/s} - 1.00 \text{ m/s}}{20.0 \text{ s}} = \boxed{-0.300 \text{ m/s}^2}$$

(b) $$\theta = \tan^{-1}\left(\frac{-0.300 \text{ m/s}^2}{0.800 \text{ m/s}^2}\right) = -20.6^\circ = \boxed{339^\circ \text{ from + X axis}}$$

(c) At $t = 25.0$ s the fish's position is specified by its coordinates and the direction of its motion is specified by the direction angle of its velocity:

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$

$$= 10.0 \text{ m} + (4.00 \text{ m/s})(25.0 \text{ s}) + \frac{1}{2}(0.800 \text{ m/s}^2)(25.0 \text{ s})^2$$

$$= \boxed{360 \text{ m}}$$

$$y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

$$= -4.00 \text{ m} + (1.00 \text{ m/s})(25.0 \text{ s}) + \frac{1}{2}(-0.300 \text{ m/s}^2)(25.0 \text{ s})^2$$

$$= \boxed{-72.7 \text{ m}}$$

$$v_{xf} = v_{xi} + a_x t = 4.00 \text{ m/s} + (0.800 \text{ m/s}^2)(25.0 \text{ s}) = 24 \text{ m/s}$$

$$v_{yf} = v_{yi} + a_y t = 1.00 \text{ m/s} - (0.300 \text{ m/s}^2)(25.0 \text{ s}) = -6.50 \text{ m/s}$$

$$\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{-6.50 \text{ m/s}}{24.0 \text{ m/s}}\right) = \boxed{-15.2^\circ}$$

12. An astronaut on a strange planet finds that she can jump a maximum horizontal distance of 15.0 m if her initial speed is 3.00 m/s. What is the free-fall acceleration on the planet?

*P4.12 From Equation 4.13 with $R = 15.0$ m, $v_i = 3.00$ m/s, and $\theta_{max} = 45.0°$:

$$g_{planet} = \frac{v_i^2 \sin 2\theta}{R} = \frac{v_i^2 \sin 90°}{R} = \frac{9.00\ \text{m}^2/\text{s}^2}{15.0\ \text{m}} = \boxed{0.600\ \text{m/s}^2}$$

15. A projectile is fired in such a way that its horizontal range is equal to three times its maximum height. What is the angle of projection?

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g};\quad R = \frac{v_i^2 (\sin 2\theta_i)}{g};\quad 3h = R$$

$$\frac{3v_i^2 \sin^2 \theta_i}{2g} = \frac{v_i^2 (\sin 2\theta_i)}{g}$$

$$\frac{2}{3} = \frac{\sin^2 \theta_i}{\sin 2\theta_i} = \frac{\tan \theta_i}{2}$$

$$\theta_i = \tan^{-1}\left(\frac{4}{3}\right) = \boxed{53.1°}$$

16. **W** To start an avalanche on a mountain slope, an artillery shell is fired with an initial velocity of 300 m/s at 55.0° above the horizontal. It explodes on the mountainside 42.0 s after firing. What are the x and y coordinates of the shell where it explodes, relative to its firing point?

The horizontal range of the projectile is found from $x = v_{xi}t = v_i \cos\theta_i t$.
Plugging in numbers,

$$x = (300\ \text{m/s})(\cos 55.0^\circ)(42.0\ \text{s})$$

$$x = \boxed{7.23 \times 10^3\ \text{m}}$$

The vertical position of the projectile is found from

$$y = v_{yi}t - \frac{1}{2}gt^2 = v_i \sin\theta_i t - \frac{1}{2}gt^2$$

Plugging in numbers,

$$y = (300\ \text{m/s})(\sin 55.0^\circ)(42.0\ \text{s}) - \frac{1}{2}(9.80\ \text{m/s}^2)(42.0\ \text{s})^2$$

$$= \boxed{1.68 \times 10^3\ \text{m}}$$

20. A ball is tossed from an upper-story window of a building. The ball is given an initial velocity of 8.00 m/s at an angle of 20.0° below the horizontal. It strikes the ground 3.00 s later. (a) How far horizontally from the base of the building does the ball strike the ground? (b) Find the height from which the ball was thrown. (c) How long does it take the ball to reach a point 10.0 m below the level of launching?

W

(a) $x_f = v_x t = (8.00\ \text{m/s})\cos 20.0^\circ\ (3.00\ \text{s}) = \boxed{22.6\ \text{m}}$

(b) Taking y positive downwards,

$$y_f = v_{yi}t + \frac{1}{2}gt^2$$

$$y_f = (8.00\ \text{m/s})\sin 20.0^\circ (3.00\ \text{s}) + \frac{1}{2}(9.80\ \text{m/s}^2)(3.00\ \text{s})^2$$

$$= \boxed{52.3\ \text{m}}$$

(c) $10.0\ \text{m} = (8.00\ \text{m/s})(\sin 20.0^\circ)t + \frac{1}{2}(9.80\ \text{m/s}^2)t^2$

$$4.90t^2 + 2.74t - 10.0 = 0$$

$$t = \frac{-2.74 \pm \sqrt{(2.74)^2 + 196}}{9.80} = \boxed{1.18\ \text{s}}$$

36. A tire 0.500 m in radius rotates at a constant rate of 200 rev/min. Find the speed and acceleration of a small stone lodged in the tread of the tire (on its outer edge).

The radius of the tire is $r = 0.500$ m. The speed of the stone on its outer edge is

$$v_t = \frac{2\pi r}{T} = \frac{2\pi(0.500\text{ m})}{(60.0\text{ s}/200\text{ rev})} = \boxed{10.5\text{ m/s}}$$

and its acceleration is

$$a = \frac{v^2}{R} = \frac{(10.5\text{ m/s})^2}{0.500\text{ m}} = \boxed{219\text{ m/s}^2\text{ inward}}$$

38. An athlete swings a ball, connected to the end of a chain, in a horizontal circle. The athlete is able to rotate the ball at the rate of 8.00 rev/s when the length of the chain is 0.600 m. When he increases the length to 0.900 m, he is able to rotate the ball only 6.00 rev/s. (a) Which rate of rotation gives the greater speed for the ball? (b) What is the centripetal acceleration of the ball at 8.00 rev/s? (c) What is the centripetal acceleration at 6.00 rev/s?

(a) Using the definition of speed and noting that the ball travels in a circular path,

$$v = \frac{d}{\Delta t} = \frac{2\pi R}{T}$$

where R is the radius of the circle and T is the period, that is, the time interval required for the ball to go around once. For the periods given in the problem,

$$8.00 \text{ rev/s} \rightarrow T = \frac{1}{8.00 \text{ rev/s}} = 0.125 \text{ s}$$

$$6.00 \text{ rev/s} \rightarrow T = \frac{1}{6.00 \text{ rev/s}} = 0.167 \text{ s}$$

Therefore, the speeds in the two cases are:

$$8.00 \text{ rev/s} \rightarrow v = \frac{2\pi(0.600 \text{ m})}{0.125 \text{ s}} = 30.2 \text{ m/s}$$

$$6.00 \text{ rev/s} \rightarrow v = \frac{2\pi(0.900 \text{ m})}{0.167 \text{ s}} = 33.9 \text{ m/s}$$

Therefore, $\boxed{6.00 \text{ rev/s}}$ gives the greater speed of the ball.

(b) Acceleration $= \frac{v^2}{r} = \frac{(9.60\pi \text{ m/s})^2}{0.600 \text{ m}} = \boxed{1.52 \times 10^2 \text{ m/s}^2}$.

(c) At 6.00 rev/s, acceleration $= \frac{(10.8\pi \text{ m/s})^2}{0.900 \text{ m}} = \boxed{1.28 \times 10^2 \text{ m/s}^2}$. So 8 rev/s gives the higher acceleration.

40. **W** Figure P4.40 represents the total acceleration of a particle moving clockwise in a circle of radius 2.50 m at a certain instant of time. For that instant, find (a) the radial acceleration of the particle, (b) the speed of the particle, and (c) its tangential acceleration.

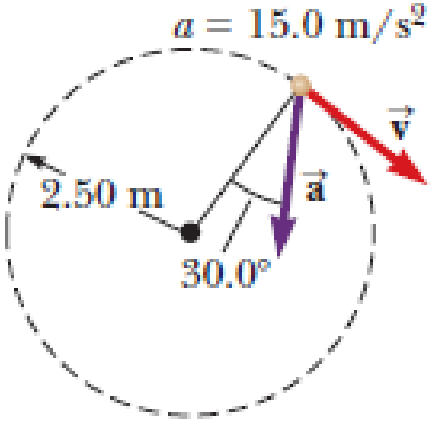


Figure P4.40

From the given magnitude and direction of the acceleration we can find both the centripetal and the tangential components. From the centripetal acceleration and radius we can find the speed in part (b). $r = 2.50$ m, $a = 15.0\ \text{m/s}^2$.



ANS. FIG. P4.40

(a) The acceleration has an inward radial component:

$$a_c = a\cos 30.0° = (15.0\ \text{m/s}^2)(\cos 30°)$$

$$= \boxed{13.0\ \text{m/s}^2}$$

(b) The speed at the instant shown can be found by using

$$a_c = \frac{v^2}{r}$$

$$v^2 = ra_c = 2.50\ \text{m}(13.0\ \text{m/s})$$

$$= 32.5\ \text{m}^2/\text{s}^2$$

$$v = \sqrt{32.5}\ \text{m/s} = \boxed{5.70\ \text{m/s}}$$

(c) $a^2 = a_t^2 + a_r^2$

so $a_t = \sqrt{a^2 - a_r^2} = \sqrt{(15.0\ \text{m/s}^2)^2 - (13.0\ \text{m/s}^2)^2} = \boxed{7.50\ \text{m/s}^2}$